

EXAMPLE 3.3

Outcomes

- Sensitivity analysis
- User defined variables using Fortran
- Column design specification

Problem Description

Using Example 3.2, plot and report the effect of varying (i) reflux ratio and (ii) distillate rate to mole purity and recovery of methanol at the distillate stream. Determine the reflux ratio and distillate rate required to achieve a separation of 99.9% recovery of methanol at distillate and 99.0% recovery of water at bottom.

1. In the Simulation pane go to Model Analysis Tool | Sensitivity sheet. Click New to create new ID, name it as S-1 and click OK to open the sensitivity analysis specification sheet as shown below.

The screenshot shows the 'Sensitivity' sheet in the Aspen Plus Model Analysis Tool. The 'Vary' tab is selected. The 'Active' checkbox is checked, and 'Case study' is unchecked. Below this, a section for 'Manipulated variables' includes a table with columns 'Variable', 'Active', 'Manipulated variable', and 'Units'. There are 'New', 'Delete', 'Copy', 'Paste', and 'Send to Aspen Multi-Case' buttons. An 'Edit selected variable' section contains a 'Manipulated variable' dropdown, a 'Type' dropdown, and 'Manipulated variable limits' radio buttons for 'Equidistant', 'Logarithmic', and 'List of values'. A 'Report labels' checkbox is at the bottom.

2. On the Vary sheet, two manipulated variables need to be defined which are reflux ratio and distillate rate. The simplest way is to use the copy and paste function. To do this, go to Block | COLUMN | Configurations sheet. Right click on the Reflux ratio value field and click Copy. Go to Vary sheet and click Paste. Repeat the same proses for Distillate rate. The two declared variables as shown below.

Vary Define Tabulate Options Cases Fortran Declarations Comments

☒ Active ☐ Case study

Manipulated variables (drag and drop variables from form to the grid below)

Variable	Active	Manipulated variable	Units
1	☒	Block-Var Block=COLUMN Variable=MOLE-RR Sent...	
2	☒	Block-Var Block=COLUMN Variable=MOLE-D Sente...	lbmol/hr

New Delete Copy Paste Send to Aspen Multi-Case

Edit selected variable

Manipulated variable

Variable: 1
Type: Block-Var
Block: COLUMN
Variable: MOLE-RR
Sentence: COL-SPECS

Manipulated variable limits

☒ Equidistant ☐ Logarithmic ☐ List of values

Start point:
End point:
☒ Number of points: 30
☐ Increment:

☒ Report labels

3. Next is to set the manipulated variable limits. For reflux ratio the limit is between 0.1 to 3. On the Manipulated variable section select Variable 1 which is the reflux ratio. On the Manipulated variables limits section enter 0.1 and 3 in the Start point and End point field respectively. Next is to set the Number of points or Increment. In this example, we want to analyse for every 0.1 increment in reflux ratio. In total 30 number of points.

Vary Define Tabulate Options Cases Fortran Declarations Comments

☒ Active ☐ Case study

Manipulated variables (drag and drop variables from form to the grid below)

Variable	Active	Manipulated variable	Units
1	☒	Block-Var Block=COLUMN Variable=MOLE-RR Sent...	
2	☒	Block-Var Block=COLUMN Variable=MOLE-D Sente...	lbmol/hr

New Delete Copy Paste Send to Aspen Multi-Case

Edit selected variable

Manipulated variable

Variable: 1
Type: Block-Var
Block: COLUMN
Variable: MOLE-RR
Sentence: COL-SPECS

Manipulated variable limits

☒ Equidistant ☐ Logarithmic ☐ List of values

Start point: 0.1
End point: 3
☐ Number of points: 30
☒ Increment: 0.1

☒ Report labels

4. For distillate rate the limit is between 1000 to 1500 lbmol/hr. On the Manipulated variable section select Variable 2 for distillate rate. On the Manipulated variables limits section enter 1000 and 1500 in the Start point and End point field respectively. The number of increment is for every 10 lbmol/hr which bring to 51 number of points.

- Next go to Define sheet to specify the sampled variables which are methanol mole purity and methanol recovery at distillate stream. Note that, samples variables are calculated values. The Copy and Paste function can be used to add sampled variables. To add methanol mole purity, go to Blocks | COLUMN | Stream Results sheet. Go to the overhead methanol mole fractions cell. Right click and select Copy. On the Define sheet click Paste to add it in the Manipulated variables list. The declared variable is shown below.

Define sheet interface showing the definition of the variable MOLEFRA. The variable is defined as Mole-Frac Stream=OVHD Substream=MIXED Component=METHANOL. The interface includes tabs for Vary, Define, Tabulate, Options, Cases, Fortran, Declarations, and Comments. Below the variable list are buttons for New, Delete, Copy, Paste, Move Up, Move Down, and View Variables. The 'Edit selected variable' section shows the variable MOLEFRA selected, with a category list (All, Blocks, Streams, Model Utility, Property Parameters, Reactions) and a reference section (Type: Mole-Frac, Stream: OVHD, Substream: MIXED, Component: METHANOL).

- For variables that are not included in the list of variables, in this case methanol recovery, it can be created using user defined parameter. Methanol recovery, R_{MeOH} , is calculated using the following equation,

$$R_{MeOH} = \frac{d_{MeOH}}{f_{MeOH}}$$

where d_{MeOH} and f_{MeOH} represent methanol distillate and feed flowrate respectively. The above equation is created using the following Fortran statement,

RECOVERY = MEOHDIST / MEOHFEED

- Three variables need to be defined. Methanol distillate (MEOHDIST) and feed flowrate (MEOHFEED) can be found in column stream results whereas recovery (RECOVERY) need to be calculated. For MEOHDIST and MEOHFEED use the Copy and Paste function to add to the sampled variables list. Rename it accordingly to MEOHDIST and MEOHFEED. The declared sampled variables are shown below.

COLUMN (RadFrac) - Stream Results (Boundary) x Main Flowsheet x Control Panel x **S-1 - Input** x +

☒ Vary ☒ Define ☐ Tabulate Options Cases Fortran Declarations Comments

Sampled variables (drag and drop variables from form to the grid below)

Variable	Definition
MOLEFRA	Mole-Frac Stream=OVHD Substream=MIXED Component=METHANOL
MEOHDIST	Mole-Flow Stream=OVHD Substream=MIXED Component=METHANOL Units=lbmol/hr
MEOHFEED	Mole-Flow Stream=FEED Substream=MIXED Component=METHANOL Units=lbmol/hr

New Delete Copy Paste Move Up Move Down View Variables

☐ Edit selected variable

8. To create a user defined parameter, in the Define sheet click New. Enter the variable name as RECOVERY and click OK. Select All in the Category section. Next, in the Reference section, for Type select Parameter. Enter 1 as Parameter no. For Physical type select Dimensionless and Unitless for Units as shown below.

COLUMN (RadFrac) - Stream Results (Boundary) x Main Flowsheet x Control Panel x **S-1 - Input** x +

☒ Vary ☒ Define ☒ Tabulate Options Cases ☒ Fortran Declarations Comments

Sampled variables (drag and drop variables from form to the grid below)

Variable	Definition
MOLEFRA	Mole-Frac Stream=OVHD Substream=MIXED Component=METHANOL
MEOHFEED	Mole-Flow Stream=FEED Substream=MIXED Component=METHANOL Units=lbmol/hr
MEOHDIST	Mole-Flow Stream=OVHD Substream=MIXED Component=METHANOL Units=lbmol/hr
RECOVERY	Parameter Parameter no.=1 Physical type=Dimensionless Units=Unitless
*	

New Delete Copy Paste Move Up Move Down View Variables

☐ Edit selected variable

Variable: **RECOVERY**

Category:

- ☒ All
- ☐ Blocks
- ☐ Streams
- ☐ Model Utility
- ☐ Property Parameters
- ☐ Reactions

Reference:

Type: **Parameter**

Parameter no.: **1**

Physical type: **Dimensionless**

Units: **Unitless**

Initial value:

9. Note that, to calculate the recovery a Fortran statement need to be created. To create a Fortran statement, go to the Fortran sheet and type the following statement. You can also use Right-click to select Variables list, then use drag-and-drop to copy variables defined on Define sheet.

Enter executable Fortran statements

```
RECOVERY = MEOHDIST / MEOHFEED
```

10. Add another variable which is water molar flowrate at distillate stream. Name it as H2ODIST. You can highlight the variable and click Move Up to rearrange the variables sequence as shown. In total there are five sampled variables.

Sampled variables (drag and drop variables from form to the grid below)

Variable	Definition
MOLEFRA	Mole-Frac Stream=OVHD Substream=MIXED Component=METHANOL
MEOHFEED	Mole-Flow Stream=FEED Substream=MIXED Component=METHANOL Units=lbmol/hr
MEOHDIST	Mole-Flow Stream=OVHD Substream=MIXED Component=METHANOL Units=lbmol/hr
H2ODIST	Mole-Flow Stream=OVHD Substream=MIXED Component=WATER Units=lbmol/hr
RECOVERY	Parameter Parameter no.=1 Physical type=Dimensionless Units=Unitless

New Delete Copy Paste Move Up Move Down View Variables

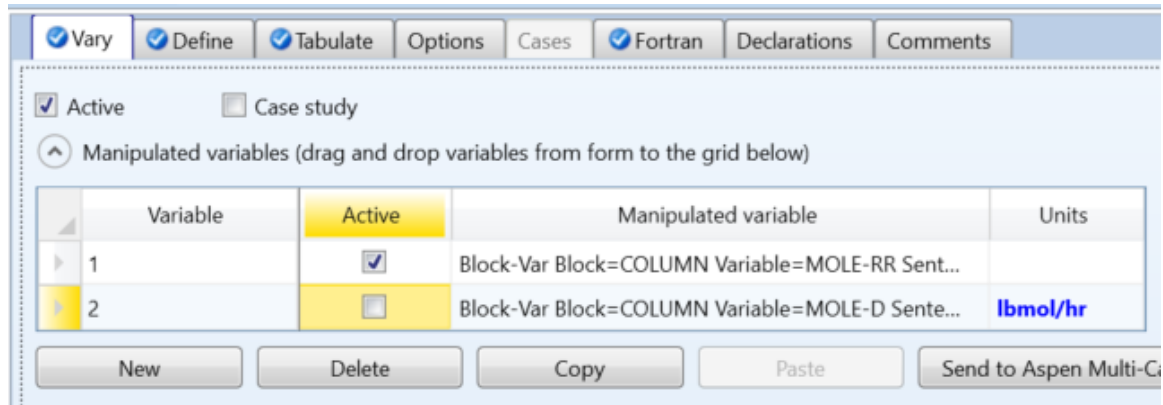
11. Next go to Tabulate sheet. Click Fill Variables to list all the variables defined in Define sheet to be plotted.

Fill Variables Table Format

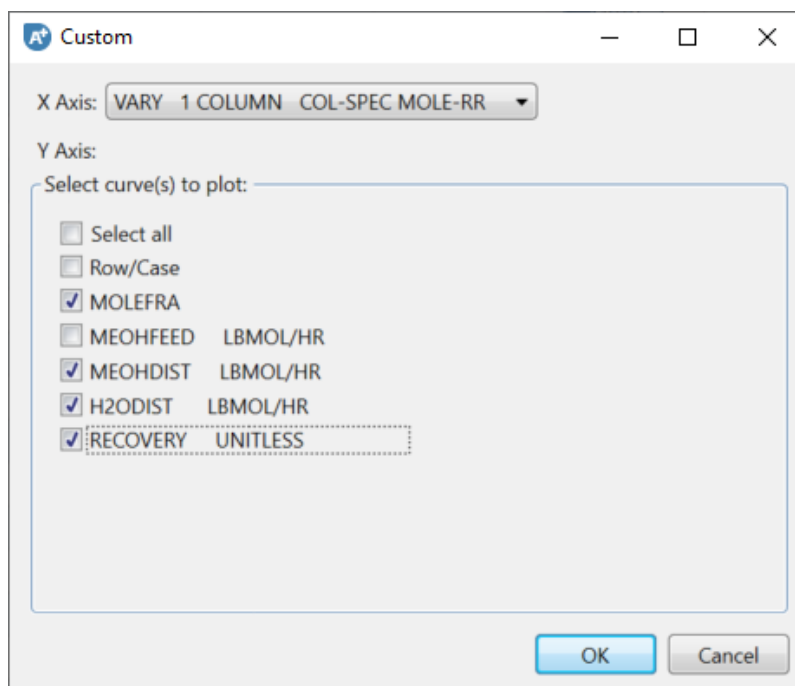
Column No.	Tabulated variable or expression
1	MOLEFRA
2	MEOHFEED
3	MEOHDIST
4	H2ODIST
5	RECOVERY
*	

↑ ↓

12. To perform the sensitivity analysis, go to Vary sheet and untick the Active button for Variable 2. Only Variable 1 will be varied. This will reduce the number of iterations. Click Run to run the sensitivity analysis. Note that, for sensitivity analysis the simulation results will not be affected.

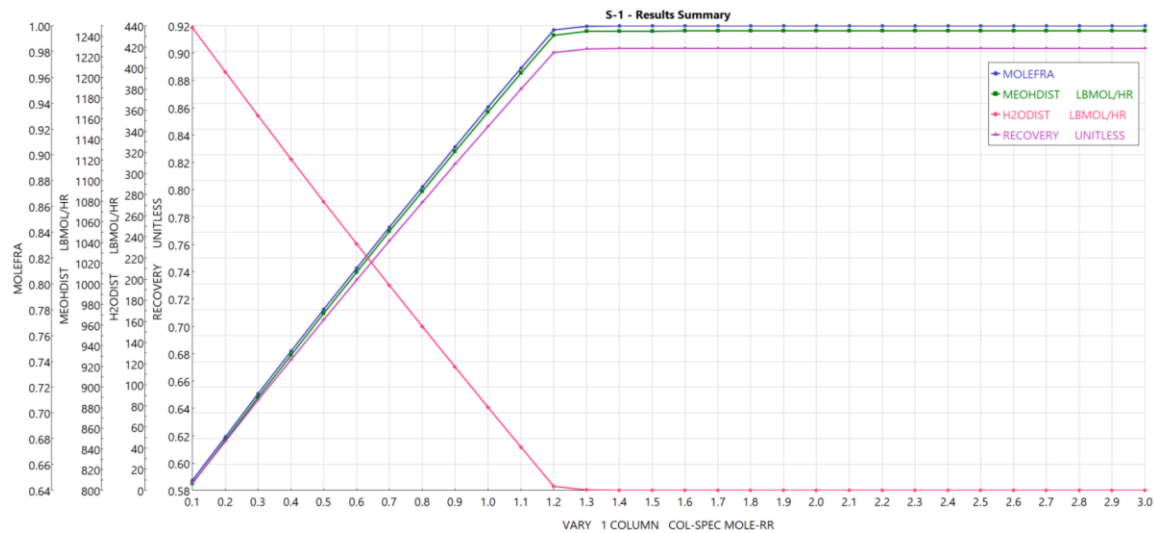


13. To view the results, go to Model Analysis Tools | Sensitivity | S-1 | Results | Summary sheet. To plot the results, go to Home ribbon. In the Plot section click Custom plot to bring up the Custom window.



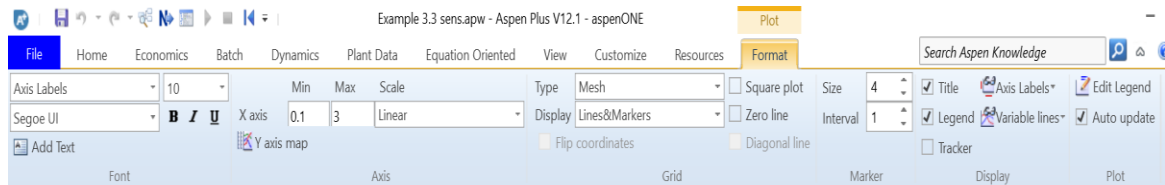
On the Custom window select X axis to VARY 1 COLUMN COL-SPEC MOLE-RR. Whereas, for Y axis select MOLEFRAC, MEOHDIST, H2ODIST and RECOVERY. Click OK and the plotted data is shown below.

INDUSTRIAL PROCESS MODELLING & SIMULATION TRAINING PART 3: DISTILLATION MODELLING & SIMULATION

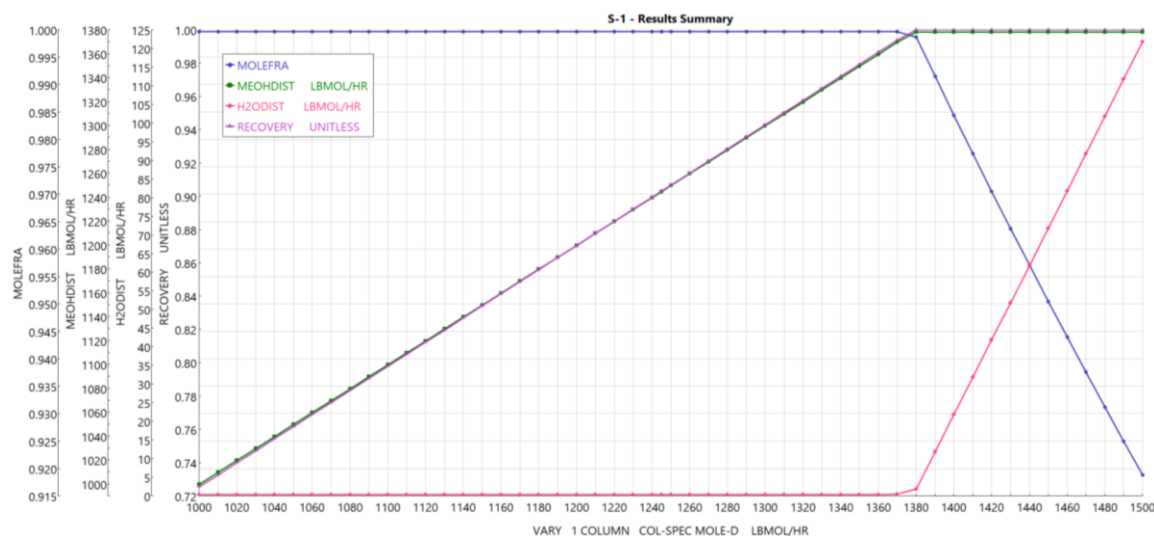


The graph shows that by increasing the reflux ratio, increases the methanol recovery and methanol flowrate. Water flowrate was observed to decreased and this in return gives high purity of methanol. This trend, however, remain constant after reflux ratio of 1.3.

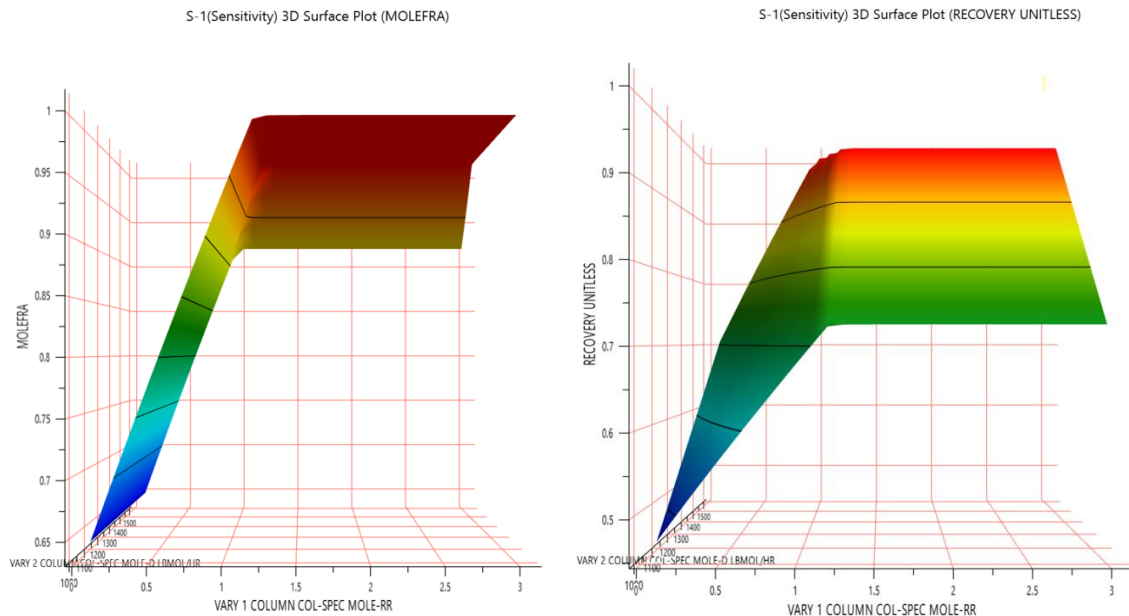
14. To change the graph labels, go to Plot | Format ribbon to adjust the graph appearance accordingly.



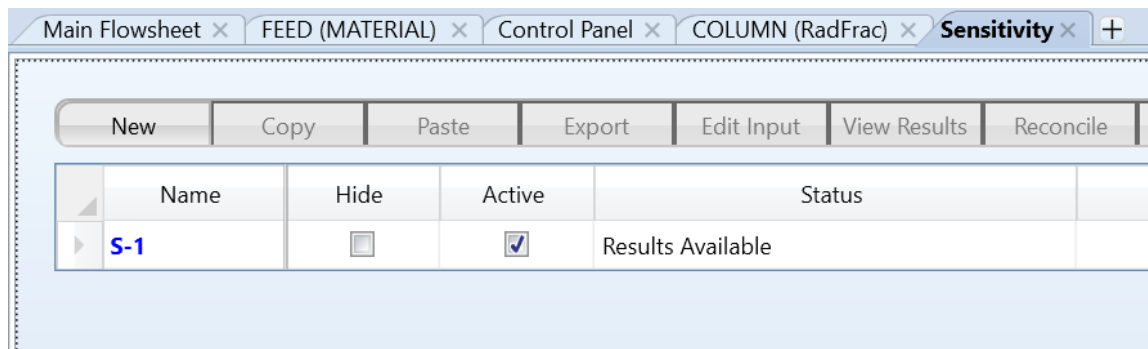
15. Repeat the same step in 11-12 but this time for Variable 2. The resulted plot is shown below.



16. Repeat the same steps 11 to 12 but for both Variable 1 and 2. Select 3D Surface Plot. On the Sensitivity 3D Surface Plot window select VARY 1 COL-SPEC MOLE-RR and VARY 1 COL-SPEC MOLE-D LBMOL/HR as Independent Variable 1 and 2, respectively. Select RECOVERY UNITLESS as Dependent Variable and click OK. Do the same for MOLEFRA. Both plots are shown below.



17. Design specifications. Before performing design specifications, it is recommended to deactivate the sensitivity analysis by unclick the Active button on the Sensitivity | S-1 sheet.



18. In this example it is desired to achieve a separation of 99.9% recovery of methanol at the distillate stream and 99.0% recovery of water at bottom by manipulating the reflux.
19. To do this, on the Navigation Pane, go to Block | COLUMN and double click on Specifications and then Design Specifications. To create a design variable, click New. On the Specifications tab, select Mole recovery and enter 0.999 as the Specification target.

The screenshot shows the 'Specifications' tab of a process simulation software. The 'Description' field contains 'Mole recovery, 0.999'. Below it, the 'Design specification' section has a 'Type' dropdown menu set to 'Mole recovery'. The 'Specification' section has a 'Target' field set to '0.999'.

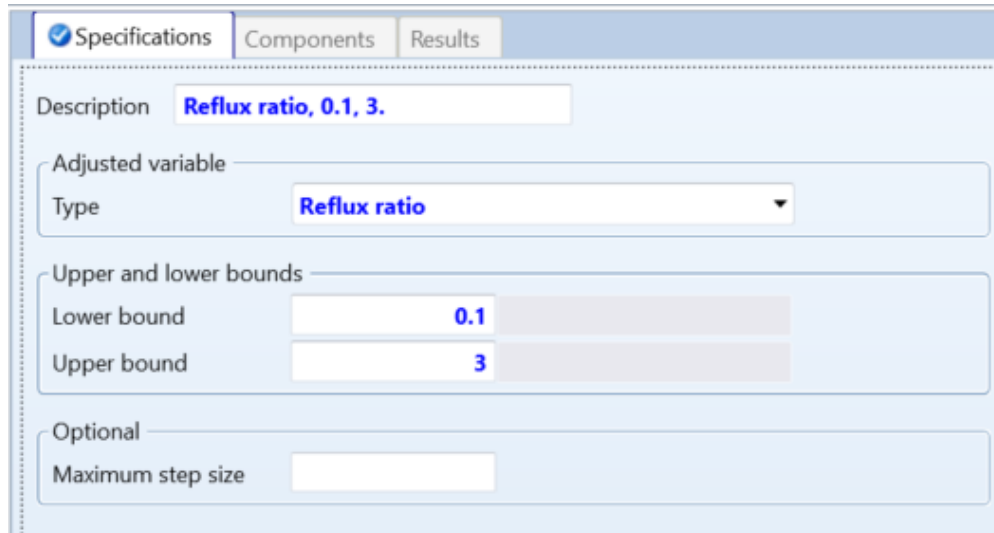
Next go to the Components tab, select methanol in the Available components list and click the > button to move the selected component into the Selected components box.

The screenshot shows the 'Components' tab of a process simulation software. It displays two lists: 'Available components' containing 'WATER' and 'Selected components' containing 'METHANOL'.

Next go to Feed/Product Streams tab. On the Product streams section select OVHD and click the > button. For the Feed streams as base streams section, select FEED and click the > button.

The screenshot shows the 'Feed/Product Streams' tab of a process simulation software. It displays two sections: 'Product streams' and 'Feed streams as base streams'. In the 'Product streams' section, 'BTMS' is selected in the left list and 'OVHD' is selected in the right list. In the 'Feed streams as base streams' section, 'FEED' is selected in the right list. Navigation buttons (>, >>, <, <<) are present between the lists in both sections.

20. Repeat the same step in 19 for 99.0% recovery of water at bottom stream.
21. Define manipulated variable. From the Navigation Pane, go to Block | COLUMN | Vary folder. Click New. On the Specifications tab, select Reflux ratio as Adjusted variable and enter 0.1 and 3 for Lower and Upper bound, respectively.



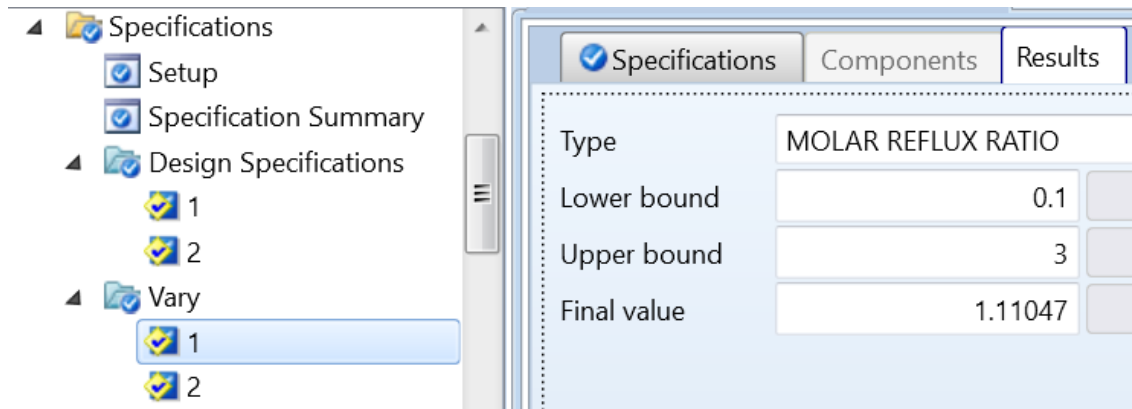
The screenshot shows the 'Specifications' tab of a software interface. The 'Description' field contains 'Reflux ratio, 0.1, 3.'. The 'Adjusted variable' section has a 'Type' dropdown menu set to 'Reflux ratio'. The 'Upper and lower bounds' section has 'Lower bound' set to 0.1 and 'Upper bound' set to 3. The 'Optional' section has a 'Maximum step size' field that is empty.

To create another variable, go to Vary and click New. Select Distillate rate as the Adjusted variable type. Enter repeat 1000 lbmol/hr and 1500 lbmol/hr for Lower and Upper bound, respectively.

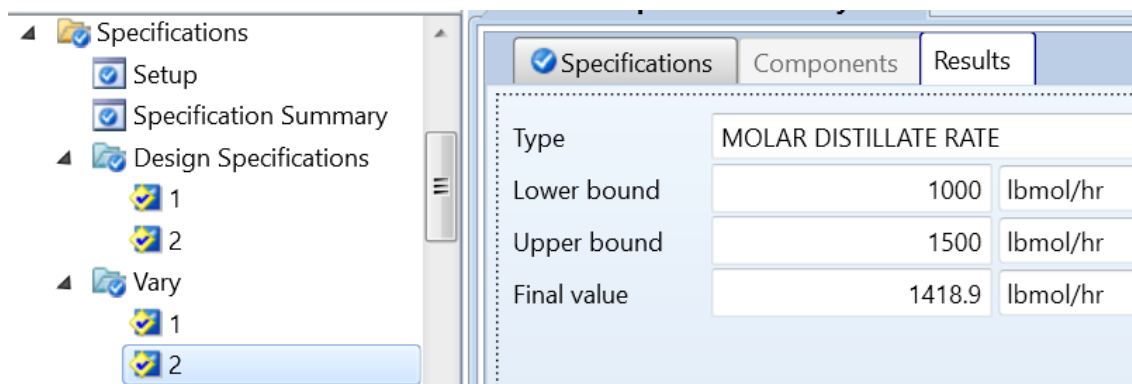


The screenshot shows the 'Specifications' tab of a software interface. The 'Description' field contains 'Distillate rate, 1000., 1500.'. The 'Adjusted variable' section has a 'Type' dropdown menu set to 'Distillate rate'. The 'Upper and lower bounds' section has 'Lower bound' set to 1000 lbmol/hr and 'Upper bound' set to 1500 lbmol/hr. The 'Optional' section has a 'Maximum step size' field that is empty.

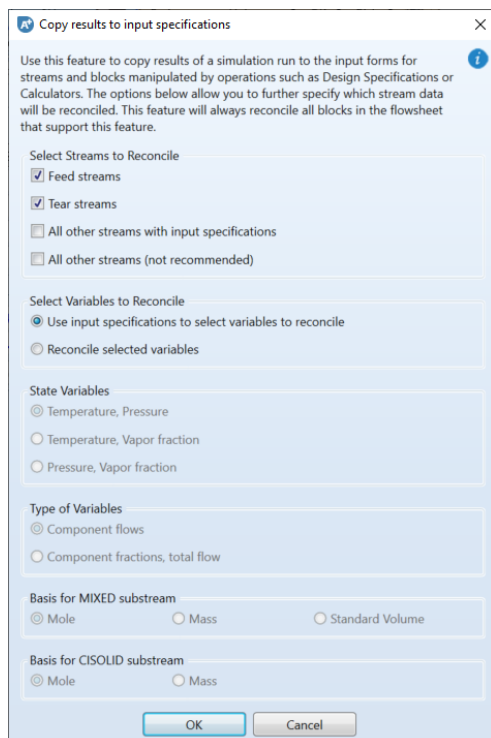
22. Click the Reset button on the Home ribbon to reinitialize the simulation. Run the simulation by pressing F5.
23. To check the results, go to Blocks | COLUMN | Specifications | Vary. To view the final value for reflux ratio, click 1 and select Results tab. The result is shown below



For distillate rate, click 1 and select Results tab to view the distillate rate final value.



24. To update the design specification results to the input specifications, click Reconcile on the Home ribbon. On the Copy results to input specifications window click OK.



25. Go to Blocks | COLUMN | Setup to view the updated input specification values. Run the simulation.

Configuration | Streams | Pressure | Condenser | Reboiler | 3-Phase | Comments

Setup options

Calculation type: *Equilibrium*

Number of stages: *40* [Stage Wizard]

Condenser: *Total*

Reboiler: *Kettle*

Valid phases: *Vapor-Liquid*

Convergence: *Standard*

Operating specifications

Distillate rate: *Mole* *1418.9* *lbmol/hr*

Reflux ratio: *Mole* *1.11047*

Free water reflux ratio: *0* [Feed Basis]

[Design and specify column internals]

26. The mole flows results are shown below. Manual calculation shows that methanol recovery at distillate and water recovery at bottom are 99.9% and 99%, respectively.

Material	Heat	Load	Vol.% Curves	Wt. % Curves	Petroleum	Polymers	Solids			
					Units			FEED	BTMS	OVHD
— Mole Flows					lbmol/hr			5587.94	4169.04	1418.9
METHANOL					lbmol/hr			1378.18	1.37833	1376.81
WATER					lbmol/hr			4209.76	4167.66	42.0977